

Aluffi, Chapter-0

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Example. aluffi Define $\mathbf{C}$ as follows: Let S be any set, and define a category $\mathbf{C} := \hat{\mathbf{S}}$ by setting

- $\text{Obj}(\hat{\mathbf{S}}) = \mathcal{P}(S)$
- for any pair of objects $A, B \in \text{Obj}(\hat{\mathbf{S}})$ let $\text{Hom}_{\mathbf{C}}(A, B)$ be the pair (A, B) if $A \subseteq B$, and let $\text{Hom}_{\mathbf{C}}(A, B) = \emptyset$ otherwise.

Indeed, notice that since every $A \in \text{Obj}(\hat{\mathbf{S}})$ satisfies $A \subseteq A$, every $\text{Hom}_{\mathbf{C}}(A, A)$ is nonempty with the identity morphism 1_A being the pair (A, A) .

Next, if there is morphism $A \rightarrow B$ and a morphism $B \rightarrow C$ then this means

that there is a morphism $A \rightarrow C$ which is just the pair (A, C) . Indeed, since $A \subseteq B$ and $B \subseteq C$ implies that $A \subseteq C$, so $\text{Hom}_{\mathbf{C}}(A, C)$ is nonempty.

Next, if there are morphisms $f : A \rightarrow B$, $g : B \rightarrow C$ and $h : C \rightarrow D$ for objects $A, B, C, D \in \text{Obj}(\hat{\mathbf{S}})$ then the morphism $hg = (B, D)$, $f = (A, B)$, $gf = (A, C)$ and so $(hg)f = (A, D) = h(gf)$. So, associativity also holds.

That $A \xrightarrow{f} B$ satisfies $f1_A = f$ and $1_B f = f$ is obvious because $1_A = (A, A)$ and $f = (A, B)$ so the morphism $f1_A$ is just (A, B) while the morphism $1_B f$ is also just (A, B) .

Example. aluffi Fix a set S and a relation $\sim$ that is reflexive and transitive. Then define a category $\mathbf{C}$ as such.

- the objects of $\mathbf{C}$ are elements of S .
- if $a, b \in \text{Obj}(\mathbf{C})$ then $\text{Hom}_{\mathbf{C}}(a, b)$ is a set containing the element (a, b) if $a \sim b$ and $\text{Hom}_{\mathbf{C}}(a, b) = \emptyset$ otherwise.

By reflexivity, $\text{Hom}(a, a)$ is always nonempty because it contains the element $1_a = (a, a)$ always.

If $f : a \rightarrow b$ and $g : b \rightarrow c$ are any two morphisms then this means that $a \sim b$ and $b \sim c$. By transitivity, $a \sim c$ and so $gf : A \rightarrow C$ is a well defined morphism. In fact it is the element (a, c) . That is, $\text{Hom}(a, c) = \{gf\} = \{(a, c)\}$.

If $f : a \rightarrow b$, $g : b \rightarrow c$ and $h : c \rightarrow d$ are morphisms in $\mathbf{C}$ then

$hg = (b, d)$ and $gf = (a, c)$ so $(hg)f = (a, d) = h(gf)$.

Next, for any morphism $f : a \rightarrow b$, $1_b f = (a, b) = f$ and $f 1_a = (a, b) = f$.

An interesting fact about this category is that given any morphism $f : a \rightarrow b$, take any object c and a pair of morphisms $\alpha', \alpha'' : c \rightarrow a$. If c, α', α'' are so that

$$f\alpha' = f\alpha''$$

then this implies that $\alpha' = \alpha''$. In fact $\alpha' = \alpha''$ is always true because $\text{Hom}_{\mathbf{C}}(c, a)$ is always a singleton or empty. This discussion elucidates the fact that every morphism in this category is a *monomorphism*.

In fact, take any morphism $f : a \rightarrow b$ and fix any object $c \in \text{Obj}(\mathbf{C})$. Take any pair of morphisms $\beta', \beta'' : b \rightarrow c$ so that $\beta'f = \beta''f$. Then, $\beta' = \beta''$.

Once again, $\beta' = \beta''$ always holds for the same reason as above. Also, $\beta'f = \beta''f$ also always holds because $\beta'f = (b, c)$ and so is $\beta''f$. Every morphism in this category is an *epimorphism*.

Example. aluffi Let $\mathbf{C}$ be a category and fix $A \in \text{Obj}(\mathbf{C})$. We will define another category from $\mathbf{C}$, denoted $\mathbf{C}_A$ as

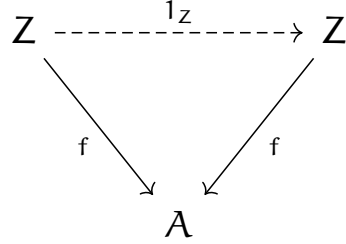
- objects of $\mathbf{C}_A$ are morphisms from objects of $\mathbf{C}$ to A . That is, elements of $\text{Obj}(\mathbf{C}_A)$ are of the form $Z \xrightarrow{f} A$ for any object $Z \in \text{Obj}(\mathbf{C})$.
- morphisms between two objects $f, g \in \text{Obj}(\mathbf{C}_A)$ are diagrams comprising of objects of $\mathbf{C}$ and the appropriate morphisms between them.

Given two objects $f : Z_1 \rightarrow A$ and $g : Z_2 \rightarrow A$ in $\mathbf{C}_A$ a morphism between f, g in $\mathbf{C}_A$ is a *commutative diagram* of the form

$$\begin{array}{ccc} Z_1 & \xrightarrow{\quad \sigma \quad} & Z_2 \\ & \searrow f & \swarrow g \\ & A & \end{array}$$

We will show that this construction of $\mathbf{C}_A$ forms a category.

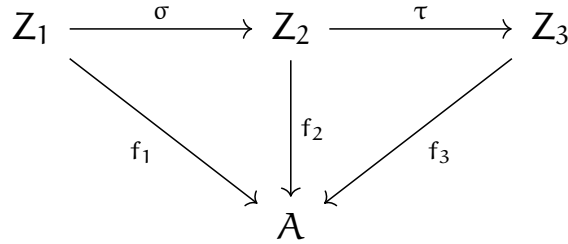
First, we will show that $\text{End}_{\mathbf{C}_A}(f)$ for any $f \in \text{Obj}(\mathbf{C}_A)$ is nonempty and contains the identity morphism by noticing that the diagram



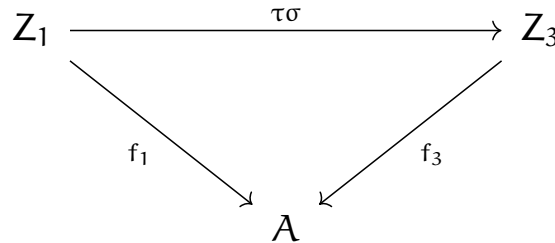
where 1_Z is the identity morphism for the object $Z \in \text{Obj}(\mathbf{C})$. This diagram is the element $1_f \in \text{Hom}_{\mathbf{C}_A}(f, f)$.

Take any two morphisms $f_1 \rightarrow f_2$ and $f_2 \rightarrow f_3$ in $\mathbf{C}_A$ next.

We will show that this defines a new morphism which is the composition of these two morphisms. For this we concatenate the commutative diagrams $f_1 \rightarrow f_2$ and $f_2 \rightarrow f_3$ side by side.



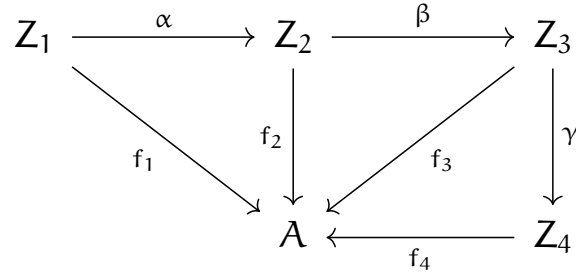
We will show that the following diagram contained in this diagram also commutes.



Indeed, since $f_1 = f_2\sigma$ and $f_2 = f_3\tau$ from the previous diagram due to commutativity so $f_1 = f_3\tau\sigma = f_3(\tau\sigma)$ which proves that this diagram also commutes, therefore, the composition of the morphisms $f_1 \rightarrow f_2$ and

$f_2 \rightarrow f_3$ is also a commutative diagram which describes the morphism $f_1 \rightarrow f_3$ between objects f_1 and f_3 of the category $\mathbf{C}_A$.

Next, suppose there are three morphisms $f_1 \rightarrow f_2 \rightarrow f_3 \rightarrow f_4$ of objects f_1, f_2, f_3 and f_4 in $\text{Obj}(\mathbf{C}_A)$. That means there are three diagrams as below

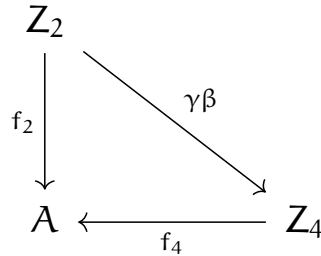


Since $f_1 \rightarrow f_2$ is a morphism of objects f_1 and f_2 in $\mathbf{C}_A$, the associated diagram above for the two morphisms is commutative. The commutativity relation is $f_1 = f_2 \alpha$. Similarly, we argue for the other three diagrams and find the following equalities by commutativity:

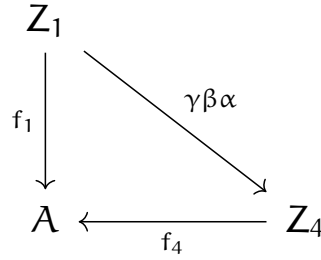
$$f_2 = f_3 \beta \text{ and}$$

$$f_3 = f_4 \gamma$$

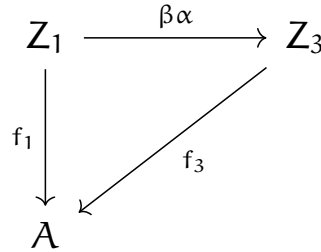
Let us set $\sigma := f_1 \rightarrow f_2, \tau := f_2 \rightarrow f_3, \delta := f_3 \rightarrow f_4$ as our diagrams. Then, the diagram $\delta \tau$ is as



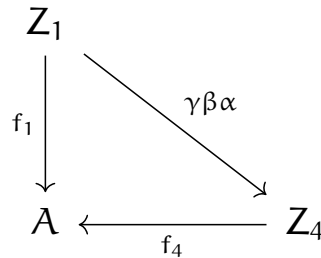
and the diagram $(\delta \tau) \sigma$ is



Next, the diagram $\tau\sigma$ is



and finally the diagram for $\delta(\tau\sigma)$ is



So, $\delta(\tau\sigma) = (\delta\tau)\sigma$ proves associativity.

That given any morphism $\alpha : f_1 \rightarrow f_2$ of C_A , $1_{f_2}\alpha = \alpha$ and $\alpha 1_{f_1} = \alpha$ is obvious from the same kinds of steps taken above.

Example. aluffi

For the category C associated to the set of integers with the $\leq$ being $\sim$ as in Eg-3.3.

fix an object $A = 3$. Then, any object of C_A is a pair $(x, 3) \in \mathbb{Z} \times \mathbb{Z}$ with $x \leq 3$. There is a morphism

$(m, 3) \rightarrow (n, 3)$ if and only if $m \leq n$.

We learned about the slice category, now let us see an example of a category that is dual to this - the coslice category.

Example 1. aluffi For a concrete example, take the category $\mathbf{Set}$ for which $\mathbf{Obj}(\mathbf{Set})$ is the class of all sets. Fix a singleton $\{*\}$ in $\mathbf{Set}$. Next, we define a new category $\mathbf{Set}^*$ whose objects are all the morphisms $f : \{*\} \rightarrow A$ for any object A of $\mathbf{Set}$. Any object in $\mathbf{Set}$ may be identified as a pair (S, s) where $s = f(*)$ because the information of any object in $\mathbf{Set}^*$ comprises of the choice of the nonempty set S and of an element $s \in S$ that f maps $*$ to in S .

A morphism between two objects (S, s) and (T, t) in $\mathbf{Set}^*$ corresponds to a map $\sigma : S \rightarrow T$ such that $\sigma(s) = t$.

Objects in $\mathbf{Set}^*$ are called *pointed sets*.

Exercise. aluffi Let $\mathbf{C}$ be a category. Consider a structure $\mathbf{C}^{\text{op}}$ with

- $\mathbf{Obj}(\mathbf{C}^{\text{op}}) := \mathbf{Obj}(\mathbf{C})$;
- for a pair of objects $A, B \in \mathbf{Obj}(\mathbf{C}^{\text{op}})$, $\text{Hom}_{\mathbf{C}^{\text{op}}}(A, B) := \text{Hom}_{\mathbf{C}}(B, A)$

that is, given any pair of objects A, B of the category $\mathbf{C}^{\text{op}}$, arrows from A to B are arrows from B to A in $\mathbf{C}$.

Solution. Take any pair of objects $A, B \in \mathbf{Obj}(\mathbf{C}^{\text{op}})$. Define

$$\odot_{\mathbf{C}^{\text{op}}} : \text{Hom}_{\mathbf{C}^{\text{op}}}(A, B) \times \text{Hom}_{\mathbf{C}^{\text{op}}}(B, C) \rightarrow \text{Hom}_{\mathbf{C}^{\text{op}}}(A, C)$$

by the following rule:

$$g \odot_{\mathbf{C}^{\text{op}}} f := f \odot g$$

where the $\odot$ in the right hand side is the composition of f and g in $\mathbf{C}$.

Since $\text{Hom}_{\mathbf{C}}(A, A)$ is nonempty for every $A \in \mathbf{Obj}(\mathbf{C})$ and objects of $\mathbf{C}^{\text{op}}$ are simply objects in $\mathbf{C}$, $\text{Hom}_{\mathbf{C}^{\text{op}}}(A, A)$ is also nonempty and contains the same $1_A \in \text{Hom}_{\mathbf{C}}(A, A)$.

The morphism $f \odot_{C^{\text{op}}} g$ is an element of $\text{Hom}_{C^{\text{op}}}(A, C)$ because $f \circ g$ is an element of $\text{Hom}_C(C, A)$. The well definedness of the map $\odot_{C^{\text{op}}}$ follows immediately from the well definedness of $\odot$. Just take elements $f, f' \in \text{Hom}_C(B, A)$ and $g, g' \in \text{Hom}_C(C, B)$ with $f = f', g = g'$ and then notice that

$$\begin{aligned} g \odot_{C^{\text{op}}} f &= f \odot g \\ &= f' \odot g' \\ &= g' \odot_{C^{\text{op}}} f' \end{aligned}$$

Next, take the morphisms $f : A \rightarrow B, g : B \rightarrow C, h : C \rightarrow D$ in C^{op} . Then,

$$\begin{aligned} (h \odot_{C^{\text{op}}} g) \odot_{C^{\text{op}}} f &= f \odot (g \odot h) \\ &= (f \odot g) \odot h \\ &= h \odot_{C^{\text{op}}} (g \odot_{C^{\text{op}}} f) \end{aligned}$$

showing that the composition is associative.

Finally, pick a morphism $f \in \text{Hom}_{C^{\text{op}}}(A, B)$. Then,

$$\begin{aligned} 1_B \odot_{C^{\text{op}}} f &= f \odot 1_B \\ &= fp \end{aligned}$$

and

$$\begin{aligned} f \odot_{C^{\text{op}}} 1_A &= 1_A \odot f \\ &= f \end{aligned}$$

This completes our construction and proof of construction of the opposite category.

Remark. We choose to indicate the morphism $f \in \text{Hom}_{C^{\text{op}}}(A, B)$ by just the same symbol when we consider it in $\text{Hom}_C(B, A)$. This might not be a bit

tricky to deal with initially but it does make writing the proof quite more convenient. ■

Exercise. aluffi If A is a finite set, how large is $\text{End}_{\text{Set}}(A)$?

Solution. The morphisms in Set are simply functions between the objects (sets) of Set . So we are simply counting the number of functions from a finite set to itself.

I claim that for a finite set A of n elements, there are n^n functions $A \rightarrow A$. Since bijections compose to give bijections, it suffices to prove the statement for a set of the form $\{1, \dots, n\}$.

We notice that there is a bijection $\{1, \dots, n\}^{\{1, \dots, n\}} \rightarrow \prod_{k=1}^n \{1, \dots, n\}$. Indeed, the map

$$f \mapsto (f(i))_{i=1}^n$$

is the obvious candidate. By the product rule of combinatorics, we see that $\prod_{k=1}^n \{1, \dots, n\}$ has cardinality n^n and we are done. ■

Exercise. aluffi Formulate precisely what it means to say that 1_a is an identity with respect to composition in Eg-3.3.

Solution. Take any pair of objects $a, b \in \text{Obj}(\mathcal{C})$. If $\text{Hom}_{\mathcal{C}}(a, b)$ is empty, then the statement is trivially true. If not then pick $(a, b) \in \text{Hom}_{\mathcal{C}}(a, b)$. It is clear that we have

$$(a, b)1_a = (a, b)(a, a) = (a, b)$$

which means that 1_a is the identity with respect to composition. We can do similar for 1_b . ■

Exercise. aluffi Can we define a category in the style of Eg-3.3 using the relation $<$ on $\mathbb{Z}$.

Solution. Every $\text{Hom}_{\mathcal{C}}(a, a)$ needs to be nonempty. If we try to define a category using $<$ we miss out on $a \sim a$ which does not give us any obvious candidate of $1_a \in \text{Hom}_{\mathcal{C}}(a, a)$. Shortly put, $<$ is not reflexive. ■

Exercise. aluffi Define in what sense the category constructed in Eg-3.4 coincides with that in Eg-3.3.

Solution. Given a set S , define the relation $\sim$ on $\mathcal{P}(S)$ as such: any two subsets A, B of S are related by $\sim$, as in $A \sim B$ if and only if $A \subseteq B$. The rest is obvious.

Simply put, the $\subseteq$ relation is a preorder. That is, $A \subseteq A$ for every subset A of S and $A \subseteq B, B \subseteq C$ means $A \subseteq C$ for any subsets A, B, C of S . ■

Exercise. aluffi A subcategory C' of a category C consists of a collection of objects of C , with morphisms $\text{Hom}_{C'}(A, B) \subseteq \text{Hom}_C(A, B)$ for all objects $A, B \in \text{Obj}(C')$, such that identities and compositions in C make C' into a category. A subcategory C' is full if $\text{Hom}_{C'}(A, B) = \text{Hom}_C(A, B)$ for all $A, B \in \text{Obj}(C')$. Construct a category of *infinite sets* and explain how it may be viewed as a full subcategory of Set .

Solution. We define a category InfSet as such:

- the objects of InfSet are simply the infinite sets in $\text{Obj}(\text{Set})$, that is $\text{Obj}(\text{InfSet}) \subseteq \text{Obj}(\text{Set})$
- given any two infinite sets $A, B \in \text{Obj}(\text{InfSet})$ we define the set of morphisms $A \rightarrow B$ by $\text{Hom}_{\text{InfSet}}(A, B) := \text{Hom}_{\text{Set}}(A, B)$.

That these constructions make InfSet into a subcategory of Set is obvious. It has the same identity and compositions as Set .

That it is a full subcategory of Set is by definition. ■

Exercise. aluffi Define a category V by taking $\text{Obj}(V) := \mathbb{N}$ and letting $\text{Hom}_V(n, m) :=$ the set of $m \times n$ matrices with real entries, for all $n, m \in \mathbb{N}$. Use product of matrices to define composition. Does this category *feel* familiar?

Solution. Given objects $j, k, l \in \text{Obj}(V)$, define morphisms $M_{k \times j} \in \text{Hom}_V(j, k)$, $N_{l \times k} \in \text{Hom}_V(k, l)$ where $M_{k \times j}$ and $N_{l \times k}$ are $k \times j$ and $l \times k$ shaped matrices of reals respectively. Define composition

$$\text{Hom}(j, k) \times \text{Hom}(k, l) \rightarrow \text{Hom}(j, l)$$

by the rule

$$M_{k \times j} \circ N_{l \times k} := N_{l \times k} M_{k \times j}$$

where the product on the right hand side is matrix multiplication which is well defined for $l \times k$ and $k \times j$ matrices. The result is a matrix $l \times j$ which is an element of $\text{Hom}_V(j, l)$.

Given any morphism $M_{n \times m}$, consider the morphism $1_n \in \text{Hom}(n, n)$. Notice that $1_n M_{n \times m} = M$ and $M_{n \times m} 1_m = M$ which shows that V is a category.

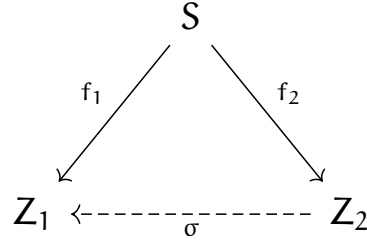
This is very familiar to the category **Vect** of all finite dimensional k -vector spaces. The objects are finite dimensional spaces and the morphisms are linear maps. The spaces can be identified by their dimension which is a natural number and the linear maps can be identified by matrices. This suggests there is a sense of *isomorphism* or *sameness* between these two categories. Such ideas have not been covered yet in the book so we will not dig further into them for now.

The matrix of 0 rows and 0 columns can represent a linear operator on the trivial vector space $\{0\}$ which has dimension 0. ■

Exercise. aluffi Define carefully objects and morphisms in Eg-3.7, and draw the diagram corresponding to composition.

Solution. Suppose C is given as a category. Then, fix some object $S \in \text{Obj}(C)$. Define the new category C^S as follows:

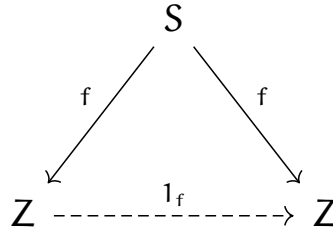
- objects in C^S are morphisms $S \rightarrow A$ for objects $A \in \text{Obj}(C)$,
- morphisms between objects f_1, f_2 in C^S are diagrams as follows:



with $f_1 = \sigma f_2$; that is to say that the diagrams are commutative.

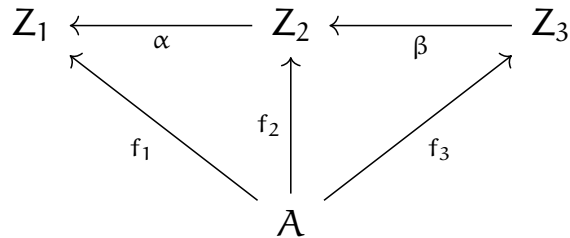
Here, $Z_1, Z_2 \in \text{Obj}(\mathbf{C})$.

Given any object $f : S \rightarrow Z \in \text{Hom}_{\mathbf{C}}(S, Z)$ of $\mathbf{C}^S$ where $Z \in \text{Obj}(\mathbf{C})$ there is a diagram



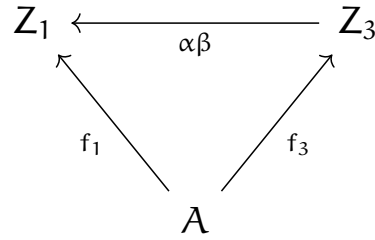
which obviously commutes and therefore, if we denote this diagram by 1_f then $1_f \in \text{Hom}_{\mathbf{C}^S}(f, f)$ for every $f \in \text{Obj}(\mathbf{C}^S)$.

Given two morphisms $f_1 \rightarrow f_2 \rightarrow f_3$ in $\mathbf{C}^S$, we will define their composition by the following commutative diagram:



This diagram indeed commutes because $f_1 = \alpha f_2$, $f_2 = \beta f_3$ so $f_1 = \alpha \beta f_3$

which gives the commutative diagram

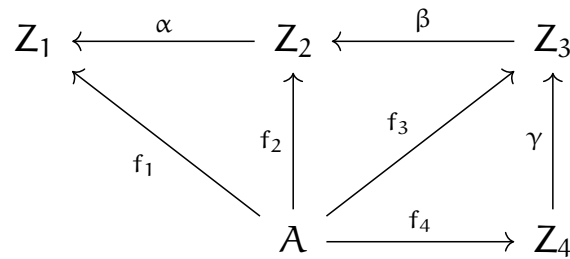


which describes a morphism $f_1 \rightarrow f_3$ proving that compositions exist and are well defined.

That 1_f is the identity with respect to composition can be proved from this easily. We shall only prove the associativity of compositions next.

For this, take three morphisms $f_1 \rightarrow f_2 \rightarrow f_3 \rightarrow f_4$ of objects f_1, f_2, f_3, f_4 in $\mathcal{C}^S$.

Consider the diagram



Next, we will label the morphisms $\sigma : f_3 \rightarrow f_4, \tau : f_2 \rightarrow f_3, \lambda : f_1 \rightarrow f_2$ and look first at $(\sigma\tau)\lambda$. It is clear that the composition $\sigma\tau$ has the diagram without Z_3 and f_3 in the middle with an arrow $Z_4 \rightarrow Z_2$ that is the morphism $\beta\gamma$. This, composed with f_1 gives us a diagram with no Z_2 and f_2 in the middle with a map $Z_4 \rightarrow Z_1$ given with the arrow $\alpha\beta\gamma$.

Next, we look at $\sigma(\tau\lambda)$. Clearly, $\tau\lambda$ gives us the diagram without Z_2 and f_2 in the middle with a morphism from $Z_3 \rightarrow Z_1$ given as $\alpha\beta$. Composing this new diagram with the diagram σ erases the f_3 and Z_3 in the middle giving an arrow $Z_4 \rightarrow Z_1$ which is the morphism $\alpha\beta\gamma$. This completes the

proof. ■

Exercise. aluffi Define a morphism between multisets, obtaining a category $\mathbf{MSet}$ containing a copy of $\mathbf{Set}$ as a full subcategory. Which objects in $\mathbf{MSet}$ determine ordinary multisets defined in 2.2 and how? What does a morphism of multiset look like here?

Solution. TODO

Exercise. aluffi Since the objects of a category $\mathbf{C}$ are not necessarily sets, it is not clear how to make sense of a notion of 'subobjects' in general. In some situations, it does make sense to talk about subobjects, and the subobjects of any given object A in $\mathbf{C}$ are in one-to-one correspondence with the morphism $A \rightarrow \Omega$ for a fixed, special object Ω of $\mathbf{C}$, called a subobject classifier. Show that $\mathbf{Set}$ has a subobject classifier. ■

Solution. Fix $\Omega := \{0, 1\} \in \mathbf{Obj}(\mathbf{Set})$. Clearly every subobject of a set A in $\mathbf{Obj}(\mathbf{Set})$ should be its subset and every subset B of A can be identified by its indicator $B \rightarrow \Omega, b \mapsto 1$ if element b of A satisfies $b \in B$ and $b \mapsto 0$ otherwise. The set of subobjects of A should be the $\mathcal{P}(A)$ so we will establish a bijection $f : \mathcal{P}(A) \rightarrow \{0, 1\}^A$.

This bijection will be given by the map $B \mapsto \mathbb{1}_B$ where $\mathbb{1}_B$ is the indicator of the set B in A . This map is injective because $\mathbb{1}_C = \mathbb{1}_B$ says an element $x \in A$ satisfies $x \in C$ if and only if $x \in B$. So, $C = B$.

For surjectivity, suppose we are given a map $g \in \{0, 1\}^A$. We will find a subset $B \subseteq A$ so that $f(B) = g$. Set $B := g^{-1}(1)$. Then, $f(B) = \mathbb{1}_B$. If we show that $\mathbb{1}_B = g$ on A then we are done. So, take any $a \in A$. If $a \in B$ then $\mathbb{1}_B(a) = 1$ and $g(a) = 1$ because $a \in g^{-1}(1)$. If $a \in A \setminus B$ then $\mathbb{1}_B(a) = 0$ while $g(a) = 0$ because $a \notin g^{-1}(1)$ and every point of A must map under g to either 0 or 1. Therefore, $g = \mathbb{1}_B$ and so $f(B) = \mathbb{1}_B = g$.

This shows that $\{0, 1\}$ is a subobject classifier for $\mathbf{Set}$.

If in a set S we equip it by the equality relation $\sim := =$ then the category $\mathbf{C}$ whose objects are $\mathbf{Obj}(\mathbf{C}) = S$ and morphisms are simply identities (a, a)

for every object $a \in S$ (all the morphisms are identities because no $a \sim b$ for distinct a and b in S). This is a category in which every morphism is an isomorphism because they're all identity. ■

Exercise. aluffi Suppose we are given n different morphisms $f_1, \dots, f_n$ so that $f_k \circ f_{k+1}$ is always well defined for $k = 1, \dots, n-1$. Then, any valid expression involving parenthesis $(,)$, elements $f_1, \dots, f_k$ and the composition operator $\circ$ is always independent of the placement of the parenthesis.

Solution. Let us denote by E_k a well formed expression of length k . By this we mean that there are k morphisms $f_1, \dots, f_k$ in E_k , $k-1$ composition operators $\circ$ and an arbitrary number of appropriately nested parenthesis.

It suffices to show that any E_k can be equivalently written in the form $(\dots (f_1 \circ f_2) \circ f_3) \circ f_4 \circ \dots) \circ f_{k-1}) \circ f_k$ which allows us to speak of the unique legal expression involving k morphisms $f_1, \dots, f_k$ and we can simply write that as $f_1 \circ f_2 \circ \dots \circ f_k$.

We proceed by strong induction on n .

For the base case of $n = 3$, $f_1 \circ (f_2 \circ f_3) = (f_1 \circ f_2) \circ f_3$ is simply the associativity of $\circ$ in C .

Our induction hypothesis is that $K_k = (\dots (f_1 \circ f_2) \circ f_3) \circ f_4 \circ \dots) \circ f_{n-1}) \circ f_n$ for all expressions K_k of length $k \in \mathbb{Z}_{>0}$ with $3 \leq k \leq n$.

Now let K_{n+1} have length $n+1$. Then, there are $l, m \in \mathbb{Z}_{>0}$ such that $l + m = n+1$ and expressions K_l and K_m such that $K_{n+1} = K_l \circ K_m$. Now, there are two cases:

Case 1: $m = 1$. Then, $l = n$, $K_m = f_{n+1}$. So, by the induction hypothesis

$$K_l = (\dots (f_1 \circ f_2) \circ \dots) \circ f_{n-1}) \circ f_n$$

Consequently,

$$K_{n+1} = (\dots (f_1 \circ f_2) \circ \dots) \circ f_{n-1}) \circ f_n) \circ f_{n+1}.$$

Case 2: $m > 1$. By the induction hypothesis, K_m can be written in the

form $K_m = K_{m-1} \circ f_{n+1}$, and so

$$K_{n+1} = K_l \circ (K_{m-1} \circ f_{n+1}) = (K_l \circ K_{m-1}) \circ f_{n+1}$$

But, $K_l \circ K_{m-1}$ is itself an expression of length n , so, by the induction hypothesis,

$$K_l \circ K_{m-1} = (\dots (f_1 \circ f_2) \circ f_3) \circ \dots \circ f_n$$

Finally,

$$K_{n+1} = (\dots (f_1 \circ f_2) \circ f_3) \circ \dots \circ f_n \circ f_{n+1}$$

which completes the induction step. ■

Exercise. aluffi Given a set S and a relation $\sim$ defined on S , define the category $\mathbf{C}$ as such: $\text{Obj}(\mathbf{C}) = S$ and given objects $a, b \in S$, if $a \sim b$ then there is a morphism $a \xrightarrow{f} b \in \text{Hom}_{\mathbf{C}}(a, b)$ given by $f = (a, b)$ and so $\text{Hom}_{\mathbf{C}}(a, b) = \{(a, b)\}$ and if a and b are not related by $\sim$ then $\text{Hom}_{\mathbf{C}}(a, b) = \emptyset$. For what conditions on $\sim$ is $\mathbf{C}$ a groupoid?

Solution. We claim that if $\sim$ is an equivalence relation then $\mathbf{C}$ is a groupoid. Indeed, we saw that $\mathbf{C}$ is a category if $\sim$ is reflexive and transitive. If $\sim$ is also symmetric, that is, if for any $a, b \in S$, $a \sim b$ then $b \sim a$. Indeed in this case, every morphism $f : a \rightarrow b$ has the left inverse $g : b \rightarrow a$ given as (b, a) so that $gf = (b, a)(a, b) = (a, a) = 1_a$. Similarly, $f : a \rightarrow b$ has the right inverse $g = (b, a)$ so that $(a, b)(b, a) = (b, b) = 1_b$. ■

Exercise. aluffi Let A, B be objects of a category $\mathbf{C}$, and let $f \in \text{Hom}_{\mathbf{C}}(A, B)$ be a morphism.

- Prove that if f has a right inverse, then f is an epimorphism.
- Find an epimorphism without a right inverse.

Solution. Suppose a morphism $f : A \rightarrow B$ between objects $A, B \in \text{Obj}(\mathbf{C})$ has a right inverse $g : B \rightarrow A$ so that $fg = 1_B$. Then, Fix an object Z and a

pair of morphisms $\alpha', \alpha'' : B \rightarrow Z$ in $\mathbf{C}$ so that $\alpha' \circ f = \alpha'' \circ f$. This implies that $\alpha' \circ f \circ g = \alpha'' \circ f \circ g$ which implies that $\alpha' \circ 1_B = \alpha'' \circ 1_B$ and therefore $\alpha' = \alpha''$.

Just take the morphism $(2, 3)$ in the category given in Eg-3.3. Any right inverse should make it so that $(2, 3) \circ g = (3, 3)$ so only $(3, 2)$ is a choice but this is not a morphism that makes sense in this category. ■

Exercise. aluffi Can you define a subcategory $\mathbf{C}_{\text{nonmono}}$ of $\mathbf{C}$ by restricting to morphisms that are *not* monomorphisms?

Solution. If $\mathbf{C}$ is the category in Eg-3.3 then every morphism is *monic*. In that case the set $\text{Hom}_{\mathbf{C}_{\text{nonmono}}}(a, a)$ for any $a \in \text{Obj}(\mathbf{C})$ is always empty because even 1_A is monic.

It might be worth thinking about whether we can do this if we do admit 1_A to every $\text{Hom}_{\mathbf{C}_{\text{nonmono}}}(a, a)$ in $\mathbf{C}_{\text{nonmono}}$. Perhaps we might have to look at some examples like $\mathbf{Grp}$ or $\mathbf{CRing}$.

Exercise. aluffi Give a concrete description of mono and epi morphisms in the category $\mathbf{MSet}$ you constructed in Ex-3.9.

Solution. TODO

References

[Paolo Aluffi(2009)]